Predicting E-Commerce Delivery Time Using Linear Regression

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Summary

This analysis aims to predict e-commerce delivery time using a linear regression model. The study focuses on the relationship between promised delivery time, regional proximity, and actual delivery time.

Analysis Code

Installing Packages

```
# Using '#' since these packages are already installed
#install.packages("tidyverse")
#install.packages("ggplot2")
#install.packages("GGally")
#install.packages("readxl")
```

Loading Libraries

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.2     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(readxl)
library(GGally)
```

```
Registered S3 method overwritten by 'GGally':
  method from
  +.gg   ggplot2
```

```r
library(ggplot2)
```

Importing the given data set by read function

```r
sales <- read_csv("sales.csv")
```

```
Rows: 10000 Columns: 12
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): order_ID, order_date, Discount Rate, order_dow
dbl (8): quantity, order_type, promise, original_unit_price, final_unit_pric...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Viewing the raw data set

```r
view(sales)
```

Data Cleaning

Now removing the missing values in actual_delivery_time column which is our dependable variable for our predictive model.

```r
sales_clean = sales |> filter(!is.na(actual_delivery_time))
view(sales_clean)
```

Rounding off actual delivery time value to refine the available data.

```r
sales_clean$actual_delivery_time <- round(sales_clean$actual_delivery_time, 2)
```

Sampling data for Analysis

To ensure efficiency, a random sample of 1000 observation is selected for our analysis

```
id = 141429531 #student ID
set.seed(id)
sales_sample <- sales_clean |> slice_sample(n = 1000)
```

Inspecting the final sample

```
view(sales_sample)
glimpse(sales_sample)
```

```
Rows: 1,000
Columns: 12
$ order_ID            <chr> "690c0820ef", "eb2ed4892d", "e456676ba7", "0148ca~
$ order_date          <chr> "3/2/2018", "3/7/2018", "3/2/2018", "3/7/2018", "~
$ quantity            <dbl> 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2~
$ order_type          <dbl> 1, 2, 2, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1~
$ promise             <dbl> 1, 3, 4, 1, 2, 1, 1, 1, 1, 1, 1, 4, 2, 4, 2, 2, 4~
$ original_unit_price <dbl> 99.0, 79.0, 69.0, 145.0, 139.0, 0.0, 59.9, 78.0, ~
$ final_unit_price    <dbl> 69.00, 39.00, 69.00, 106.00, 76.50, 0.00, 39.90, ~
$ `Discount Rate`     <chr> "30%", "51%", "0%", "27%", "45%", NA, "33%", "27%~
$ gift_item           <dbl> 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0~
$ same_region         <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0~
$ actual_delivery_time <dbl> 0.78, 5.56, 4.22, 3.90, 0.98, 1.19, 1.06, 1.02, 1~
$ order_dow           <chr> "Friday", "Wednesday", "Friday", "Wednesday", "Su~
```

```
summary(sales_sample)
```

```
   order_ID          order_date          quantity         order_type
 Length:1000        Length:1000        Min.   : 1.000   Min.   :1.000
 Class :character   Class :character   1st Qu.: 1.000   1st Qu.:1.000
 Mode  :character   Mode  :character   Median : 1.000   Median :1.000
                                       Mean   : 1.188   Mean   :1.165
                                       3rd Qu.: 1.000   3rd Qu.:1.000
                                       Max.   :24.000   Max.   :2.000
    promise      original_unit_price final_unit_price Discount Rate
 Min.   :1.000   Min.   :  0.0       Min.   : -8.00   Length:1000
 1st Qu.:1.000   1st Qu.: 65.0       1st Qu.: 39.90   Class :character
 Median :2.000   Median : 85.0       Median : 61.95   Mode  :character
```

```
 Mean   :1.946   Mean   :117.9       Mean   : 82.75
 3rd Qu.:2.000   3rd Qu.:149.0       3rd Qu.:109.00
 Max.   :8.000   Max.   :520.0       Max.   :400.00
   gift_item      same_region    actual_delivery_time  order_dow
 Min.   :0.000   Min.   :0.000   Min.   : 0.2200      Length:1000
 1st Qu.:0.000   1st Qu.:1.000   1st Qu.: 0.7475      Class :character
 Median :0.000   Median :1.000   Median : 1.0000      Mode  :character
 Mean   :0.098   Mean   :0.933   Mean   : 1.4762
 3rd Qu.:0.000   3rd Qu.:1.000   3rd Qu.: 1.8700
 Max.   :1.000   Max.   :1.000   Max.   :13.8400
```

Choosing Variables

We select *actual_delivery_time* as the dependent variable, with *promise* and *same_region* as independent variables.

Exploratory Analysis

Correlation between actual and promised delivery time

To check correlation of 2 chosen numerical variables for our analysis,

```
cor(sales_sample$actual_delivery_time,sales_sample$promise)
```

```
[1] 0.7454382
```

The returned correlation value is

`0.7208618`

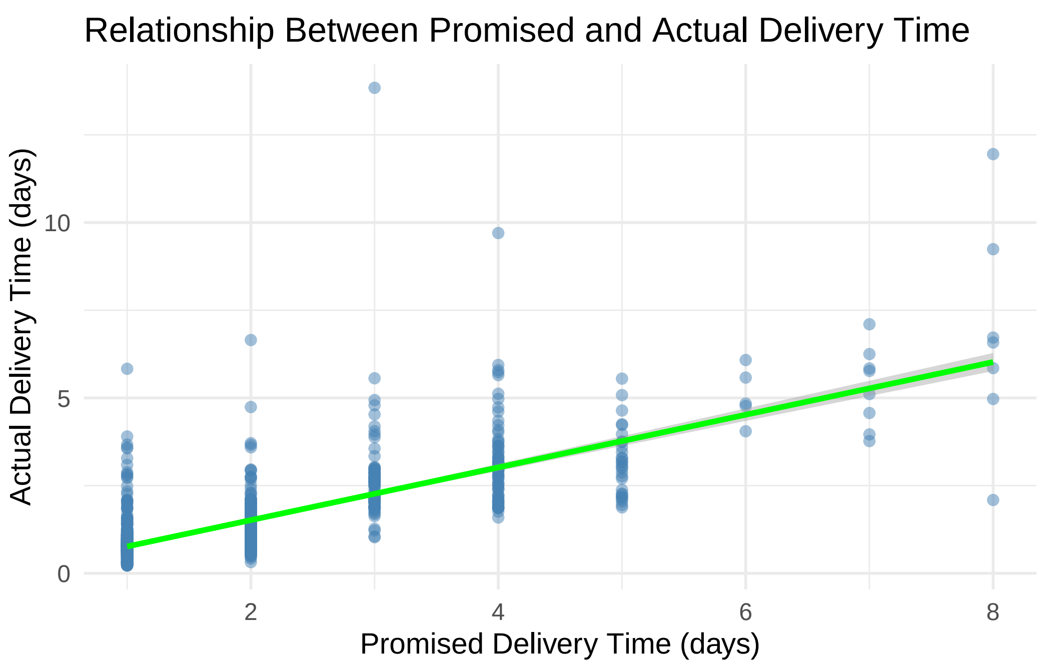
Which indicates a **strong positive linear relationship,** and it is evident that whenever promised delivery time increases, the actual delivery time also increase.

Visualizing the relationship

Also to check this correlation relationship in a plot, we will use scatter plot with regression line as below

```r
ggplot(sales_sample, aes(x = promise, y = actual_delivery_time)) +
  geom_point(alpha = 0.5, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "Green") +
  labs(title = "Relationship Between Promised and Actual Delivery Time",
       x = "Promised Delivery Time (days)",
       y = "Actual Delivery Time (days)") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Linear Regression Model

Linear regression model fitting as below:-

```r
LM <-lm(actual_delivery_time ~ promise + same_region, data = sales_sample)
```

```r
summary(LM)
```

```
Call:
```

```
lm(formula = actual_delivery_time ~ promise + same_region, data = sales_sample)

Residuals:
    Min      1Q  Median      3Q     Max 
-4.1504 -0.4364 -0.0759  0.2141 11.5100 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.45760    0.13304   -3.44 0.000607 ***
promise      0.78208    0.02262   34.57  < 2e-16 ***
same_region  0.44140    0.11556    3.82 0.000142 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8528 on 997 degrees of freedom
Multiple R-squared:  0.5621,	Adjusted R-squared:  0.5612 
F-statistic: 639.9 on 2 and 997 DF,  p-value: < 2.2e-16
```

R-Square value is **52.2%,** which is the variation in actual delivery time,

Creating model for the analysis of predicting delivery time

- Intercept (Actual_delivery time)
- Regression Coefficient (Promise)
- Regression Coefficient (Same Region)

$\hat{y} = \quad + \quad \times$ promise $+ \quad \times$ same_region

The regression equation is **actual_delivery_time = -0.46243 + 0.85531 (Promise) + 0.31589 (Region).**

The above coefficient values are stored in environment.

```
b0 <- LM$coefficients[["(Intercept)"]]
b1 <- LM$coefficients[["promise"]]
b2 <- LM$coefficients[["same_region"]]
```

Example 1 Prediction: If an order has promised delivery time of 5 days, and is in same region. Then the predicted delivery time as per the created model will be

Delivery time = -0.46243+0.85531(5)+0.31589(1) ~= 4.13 days

```
b0+b1*5+b2*1
```

```
[1] 3.894189
```

Example 2 Prediction: If an order has promised delivery time of 2 days, and is in different region. Then the predicted delivery time as per the created model will be

Delivery time = -0.46243+0.85531(2)+0.31589(0) ~= 1.24 days

```
b0+b1*2+b2*0
```

```
[1] 1.106558
```

Model Application on data set

Create new column 'predicted_delivery' using mutate

```
sales_sample <- sales_sample |>
  mutate(predicted_delivery = round(b0 + b1 * promise + b2 * same_region, 2))
```

Additional Insight

In logistics, *order_type* is an important variable. However, when assessed using a linear regression model, its statistical significance was low. Despite this, a noticeable difference in delivery time exists between direct (1P) and third-party (3P) orders.

To evaluate this, the data was grouped by *order_type*, and the mean *actual_delivery_time* for each category was calculated as follows:

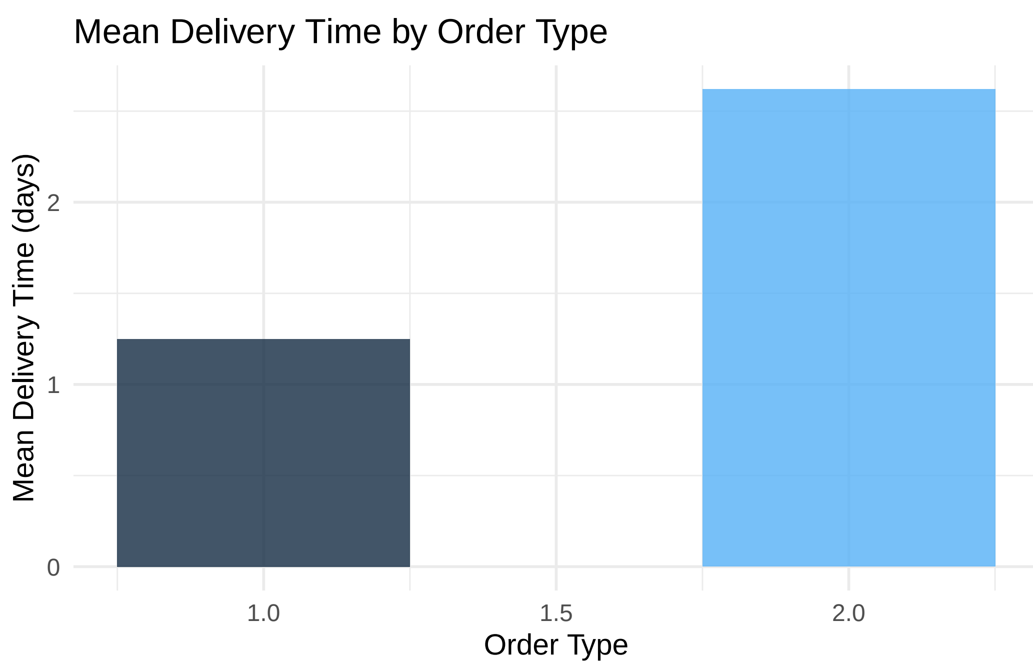
```
sales_sample |> group_by(order_type) |>
  summarise(avg_delivery = round(mean(actual_delivery_time), 2))
```

```
# A tibble: 2 x 2
  order_type avg_delivery
       <dbl>        <dbl>
1          1         1.25
2          2         2.62
```

Visualizing Delivery Time by Order Type

Creating column chart using group by and summarise function to visualize the above mean.

```
sales_sample |>
  group_by(order_type) |>
  summarise(mean_delivery = mean(actual_delivery_time, na.rm = TRUE)) |>
  ggplot(aes(x = order_type, y = mean_delivery, fill = order_type)) +
  geom_col(width = 0.5, alpha = 0.8) +
  labs(title = "Mean Delivery Time by Order Type",
       x = "Order Type",
       y = "Mean Delivery Time (days)") +
  theme_minimal() +
  theme(legend.position = "none") #ref, Gen AI to remove legend
```



It is clear that, **1st party direct orders are delivered in ~1 day**, while **3rd party orders take ~3 days**, suggesting operational differences to be addressed.

Analysis Findings & interpretation

Used linear regression to predict actual e-commerce delivery time based on promised delivery days and region.

Strong positive correlation was found between promised and actual delivery time, and same-region deliveries are generally faster.

Additional insights reveal that 1st party orders are fulfilled in ~1 day, while 3rd party orders take ~3 days on average.

These findings help the company to predict delivery time based on model and identify key factors affecting delivery performance, aiding in logistics planning thereby contributes to service improvement.